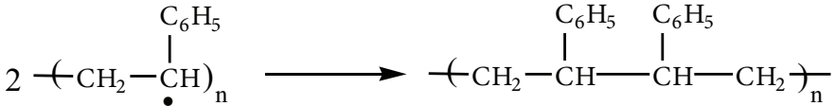
Termination

The above chain reaction can be stopped by stopping the supply of monomer or by coupling of two chains or reaction with an impurity such as oxygen.



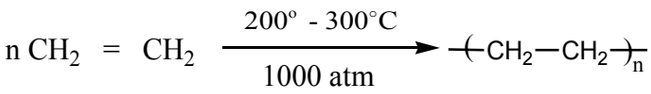
15.4. 3 Preparation of some important addition polymers

1. Polythene

It is an addition polymer of ethene. There are two types of polyethylene i) HDPE (High Density Polyethylene) ii) LDPE (Low Density polyethylene).

LDPE

It is formed by heating ethene at 200° to 300° C under oxygen as a catalyst. The reaction follows free radical mechanism. The peroxides formed from oxygen acts as a free radical initiator.



ethene Polythene

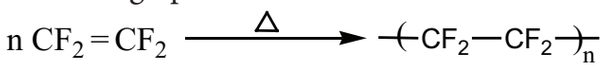
It is used as insulation for cables, making toys etc…

HDPE

The polymerization of ethylene is carried out at 373K and 6to7 atm pressure using Zeiglar – Natta catalyst $[TiCl_4+(C_2H_5)_3Al]$ HDPE has high density and melting point and it is used to make bottles, pipe etc..,

Preparation of Teflon (PTFE)

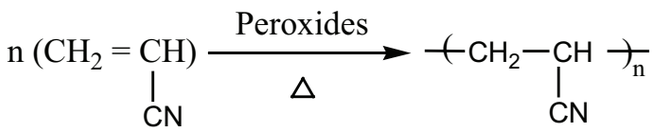
The monomer is tetrafluroethylene. When the monomer is heated with oxygen (or) ammonium persulphate under high pressure, Teflon is obtained.



It is used for coating articles and preparing non – stick utensils.

I. Preparation of Orlon (polyacrylonitrile – PAN)

It is prepared by the addition polymerisation of vinylcyanide (acrylonitrile) using a peroxide initiator.



Prop - 2-enenitrile PAN

It is used as a substitute of wool for making blankets, sweaters etc…

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